🥇 Olympics Data Analysis

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01 | About the Dataset

271,116	201	120	66
Athlete Records	Countries / NOCs	Years of History	Olympic Sports

Athlete Events Dataset

- athlete_events.csv
- 271,116 rows × 15 columns
- Coverage: 1896 → 2016
- Fields: Name, Sex, Age, Height, Weight
- Team, NOC, Year, Season, Sport, Event, Medal
- 230 unique NOCs | 66 Sports | 2 Seasons

Country Economics Dataset

- country_economics.csv
- 201 rows × 4 columns
- Fields: Country, Code, Population, GDP per Capita
- 12.4% missing GDP values
- 2.5% missing population values
- 8 territory entries (flagged with *)

Key Derived Variables: BMI, Won_Medal (0/1), GDP_Tier (Low/Mid/High), Log_Population, Medal_Rate

02 | Purpose of Analysis

Understand Performance Drivers

Identify which economic, demographic & athlete-level factors correlate with Olympic medal success across nations.

Uncover Historical Patterns

Trace how Olympic participation, gender diversity and medal dominance evolved from 1896 to 2016.

Country-Level Benchmarking

Compare nations on medal efficiency (medals per athlete) to separate resource advantages from systemic sports development.

India-Specific Insights

Understand India's Olympic underperformance relative to its population size and growing GDP — and identify gaps.

Policy-Relevant Outcomes

Generate actionable insights for policymakers, sports federations and corporate sponsors to improve athlete outcomes.

Data Quality Demonstration

Showcase best practices in data cleaning, feature engineering and exploratory analysis on real-world messy datasets.

03 | Data Cleaning Steps

1. **Territory Flagging & Strip**
   Detected 8 entries with asterisks (*) indicating territories. Created Is_Territory binary flag, then removed asterisks from country names.

2. **Population Imputation**
   5 missing population values imputed using the overall dataset median (7,450,124) — preserving central tendency without distortion.

3. **GDP Group-Median Imputation**
   25 missing GDP values filled using group-specific medians: $5,232 for countries, $42,328 for territories — accounting for structural differences.

4. **Duplicate Removal**
   1,385 exact duplicate athlete rows removed from athlete_events.csv. Dataset reduced from 271,116 → 269,731 rows.

5. **Medal NaN → 'No Medal'**
   All null Medal values replaced with 'No Medal' string — enabling category-level analysis including non-winners.

6. **Numerical Imputation (Age/Height/Weight)**
   Median imputation: Age → 24, Height → 175cm, Weight → 70kg. IQR outlier detection flagged 11,059 age outliers (range: 10–97), retained for authenticity.

7. **Feature Engineering**
   Created: BMI, Won_Medal (0/1), GDP_Tier (4 bins), Log_Population, Medal_Rate (medals/athletes) for richer analysis.

04 | Data Analysis — Steps & Approach

A

Descriptive Statistics

Computed mean, median, std for Age, Height, Weight, BMI by Season & Gender.

B

Distribution Analysis

Histograms with mean/median lines for all physical attributes. BMI trimmed to 15–50.

C

Medal Breakdown

Frequency counts and pie chart of Gold/Silver/Bronze distribution across all athletes.

D

Sports & Gender

Top 20 sports by entries; gender proportion stacked bar for top 15 sports.

E

Time-Series Analysis

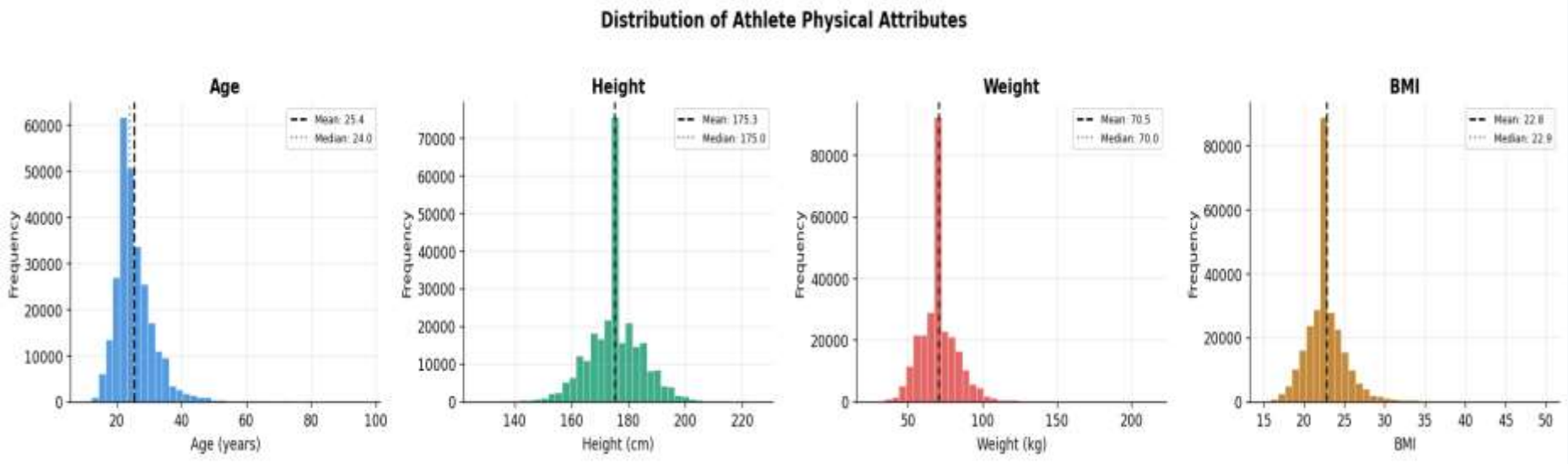
Gender trends 1896–2016; medal dominance over time for top 6 nations; participation growth.

F

Economics × Performance

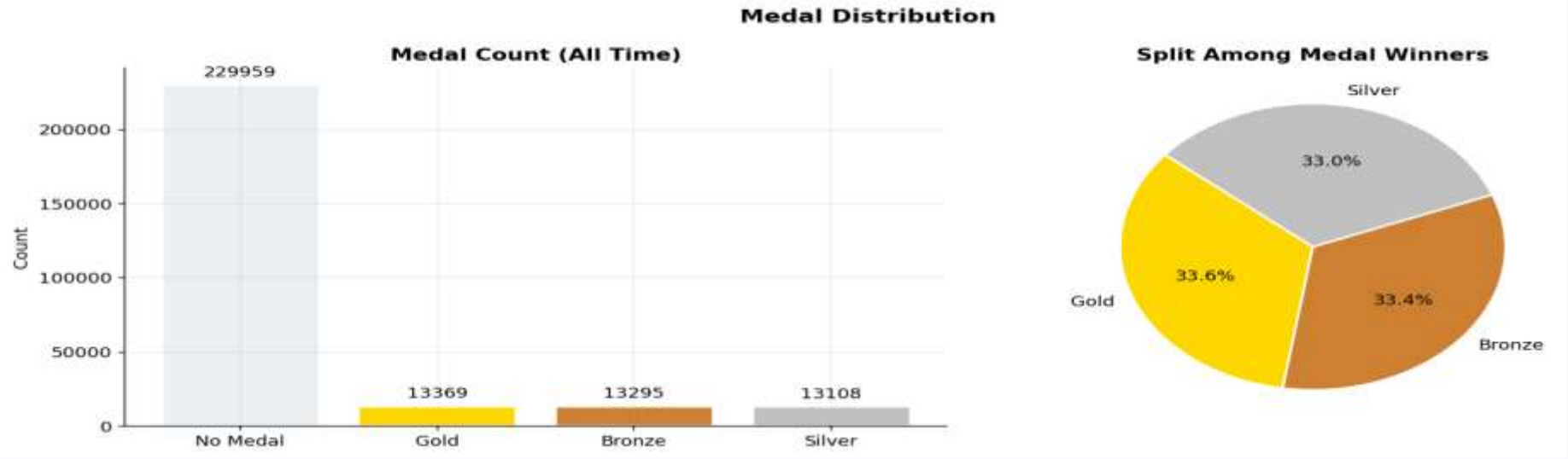
Merged athlete data with country GDP/Population. Bubble chart: GDP per capita vs total medals.

05 | Athlete Physical Attribute Distributions



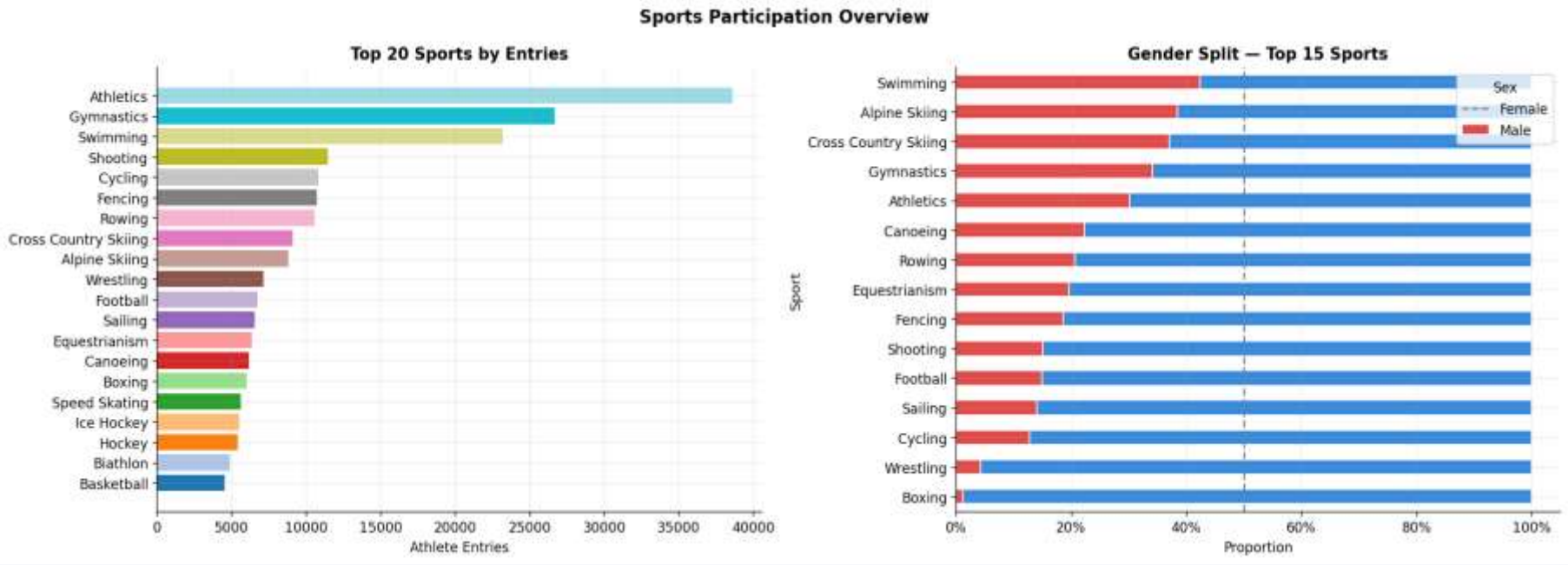
- Mean age ≈ 25.5 years; right-skewed due to veteran Olympians aged 60–97
- Height: roughly normal at ~175cm mean; Weight: slight right skew (~70kg mean)
- BMI peaks at 22–24 — typical healthy athletic range; outliers from heavy sports

05 | Medal Distribution (All-Time)



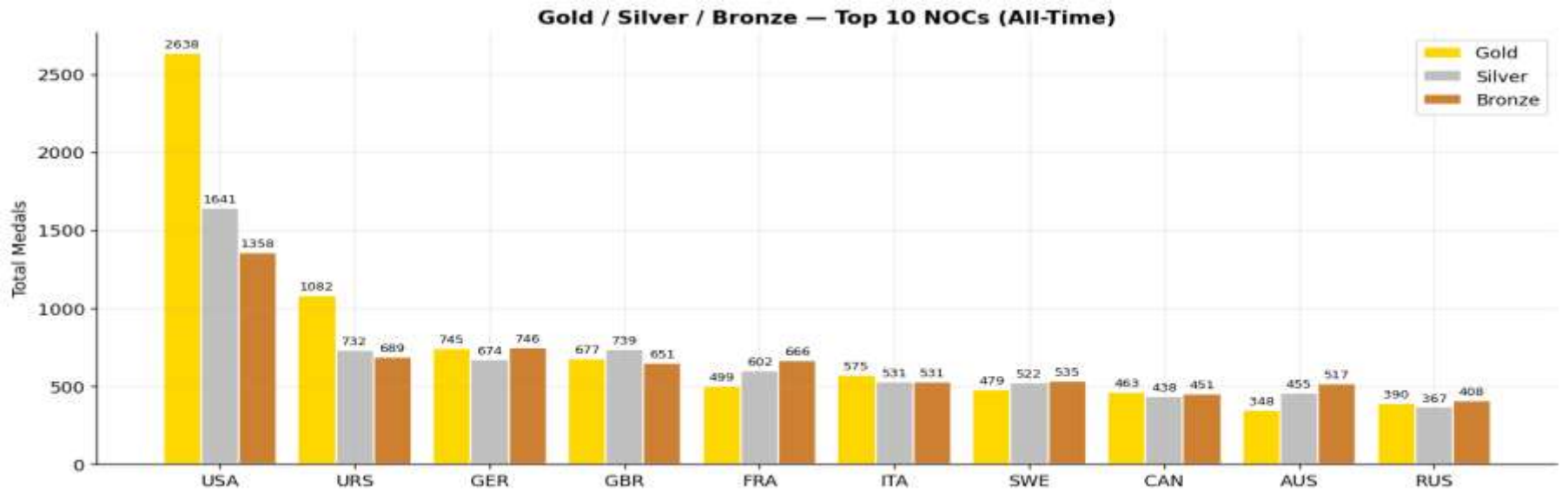
- Gold, Silver & Bronze are nearly equally distributed (~33% each) among medal winners
- Vast majority of athlete-entries (>85%) receive no medal — reflecting high competition

05 | Sports Participation & Gender Split



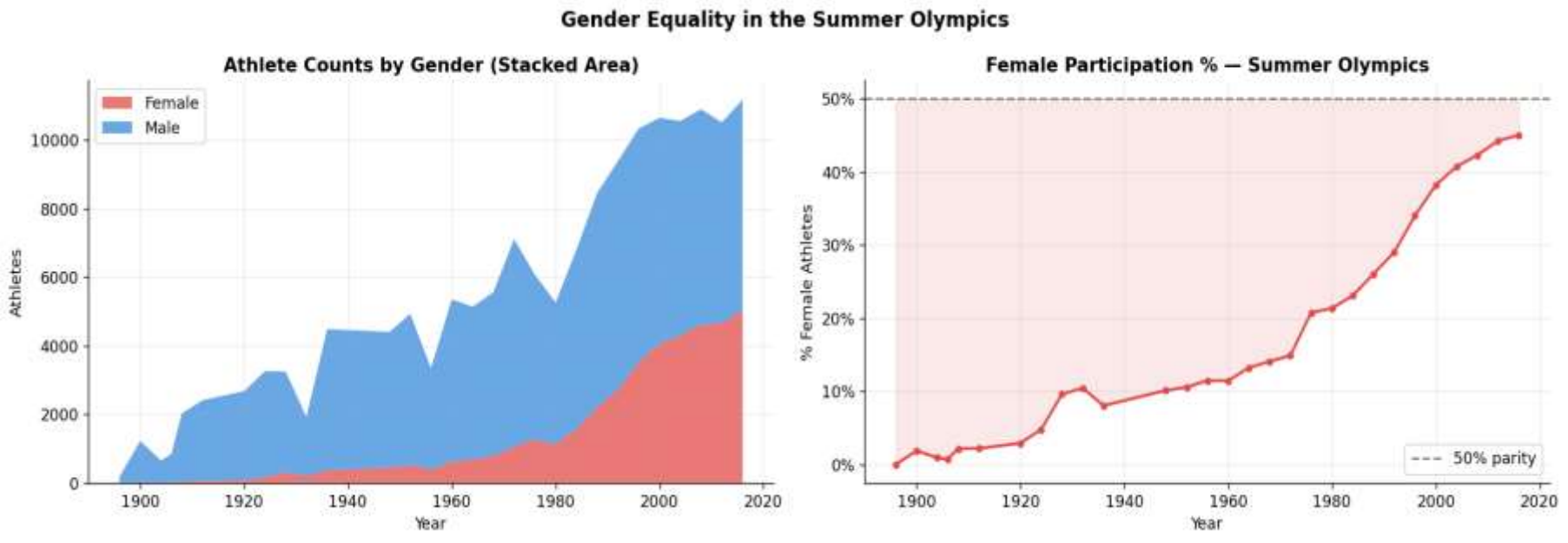
- Athletics, Gymnastics & Swimming dominate total athlete entries by a large margin
- Women's share in sports grew significantly post-1984 reforms

05 | Medal Breakdown: Top 10 Nations (All-Time)



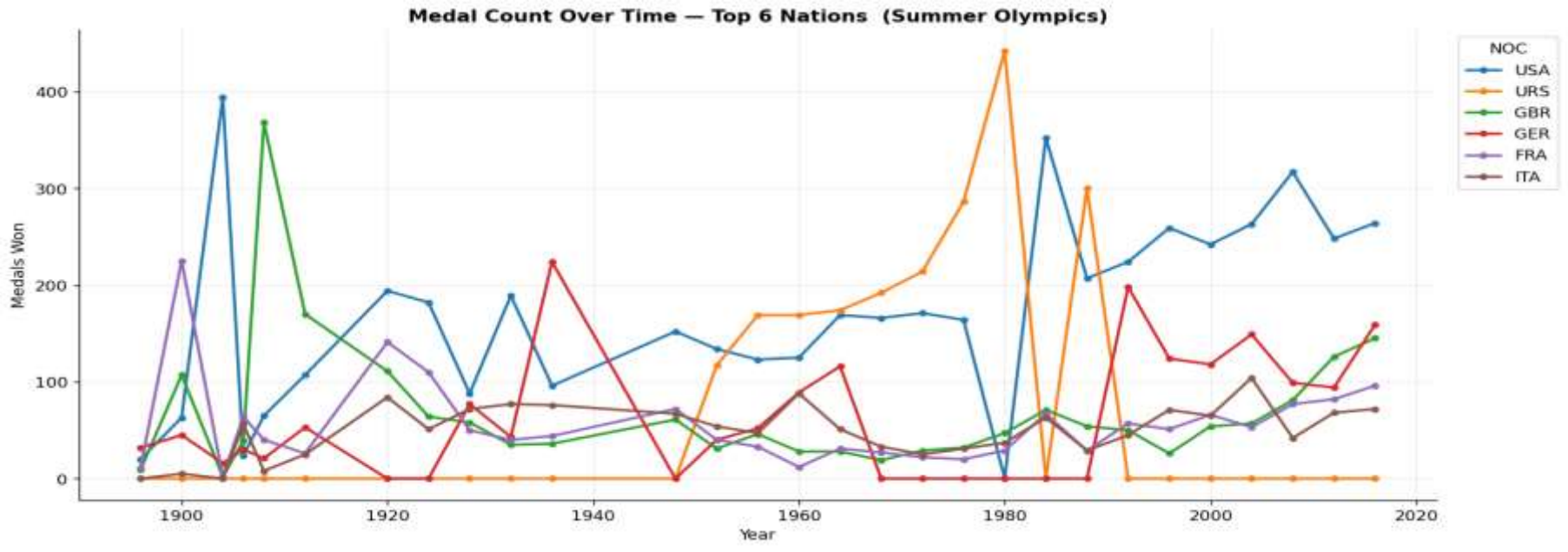
- USA leads all-time with the most Gold medals.
- China (CHN) entered the top 10 only after 1984, showing rapid sports development
- India (IND) does not appear in this top 10 — a major gap for a country of its population

05 | Gender Equality in the Summer Olympics



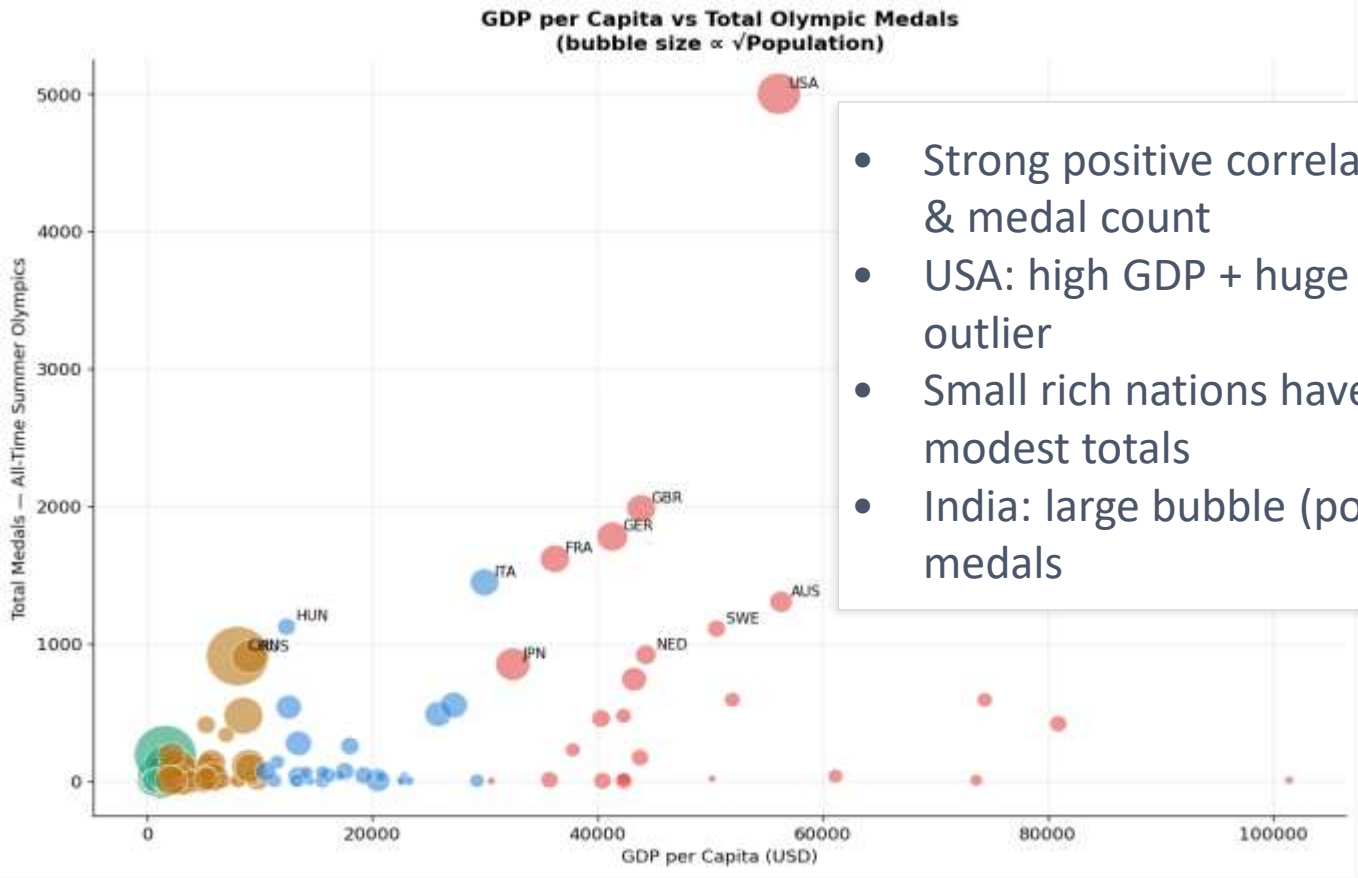
- Women were virtually absent until 1920s; IOC inclusion reforms accelerated from 1970s onward
- Female athletes reached ~45% of total Summer Olympics participants by 2016.

05 | Medal Dominance: Top 6 Nations



- Post-1992: China's rapid rise in Summer medals reflects its state-led sports investment model
- Dip in 1980 (USA boycott) & 1984 (Soviet boycott) visible in the trends

05 | Economics × Performance



- Strong positive correlation between GDP/capita & medal count
- USA: high GDP + huge population = top-right outlier
- Small rich nations have high medal rate but modest totals
- India: large bubble (population) but very low medals

06 | India's Olympic Performance vs Global Peers

Country	Total Medals	Gold Medals	Medal Rate	GDP/Capita
USA US	5,637	2,638	0.31	$55,000+
China CN	1,395	596	0.25	$8,000+
Russia/URS RU	3,947	1,122	0.28	$12,000+
Australia AU	1,344	503	0.29	$50,000+
India IN	**35**	**10**	**0.006**	**$2,000 (then)**
Kenya KE	107	34	0.18	$1,400
Jamaica JM	78	26	0.27	$5,000

35
Total Medals
(1900–2016)

10
Gold Medals
(All-time)

0.006
Medal Rate
(medals/athlete)

#1.4B
Population
(World's largest)

⚠ India's medal rate (0.006) is among the lowest globally — Kenya and Jamaica, with far smaller populations & lower GDPs, outperform India significantly.

06 | India — Sport-Wise Medal Breakdown

Field Hockey	**Dominant (8 Golds)**	**25**
Wrestling	3	Recent (KD Jadhav, Sushil, Bajrang)
Shooting	3	Recent (Manu Bhaker, Sarabjot Singh, Swapnil Kusale)
Boxing	1	Mary Kom — 2012
Athletics	1	Norman Pritchard — 1900
Badminton	1	Saina/Sindhu — 2012/16

⚠ 71% of India's all-time medals come from a single sport — Field Hockey. Post-1980, India's hockey dominance collapsed, exposing deep structural dependence.

06 | What This Analysis Tells Us About India

Q: Is India's poor performance about poverty?
A: Partially. India's historical GDP/capita was low, but nations like Jamaica ($5K GDP) and Kenya ($1.4K GDP) vastly outperform India per capita. Economics alone cannot explain the gap.

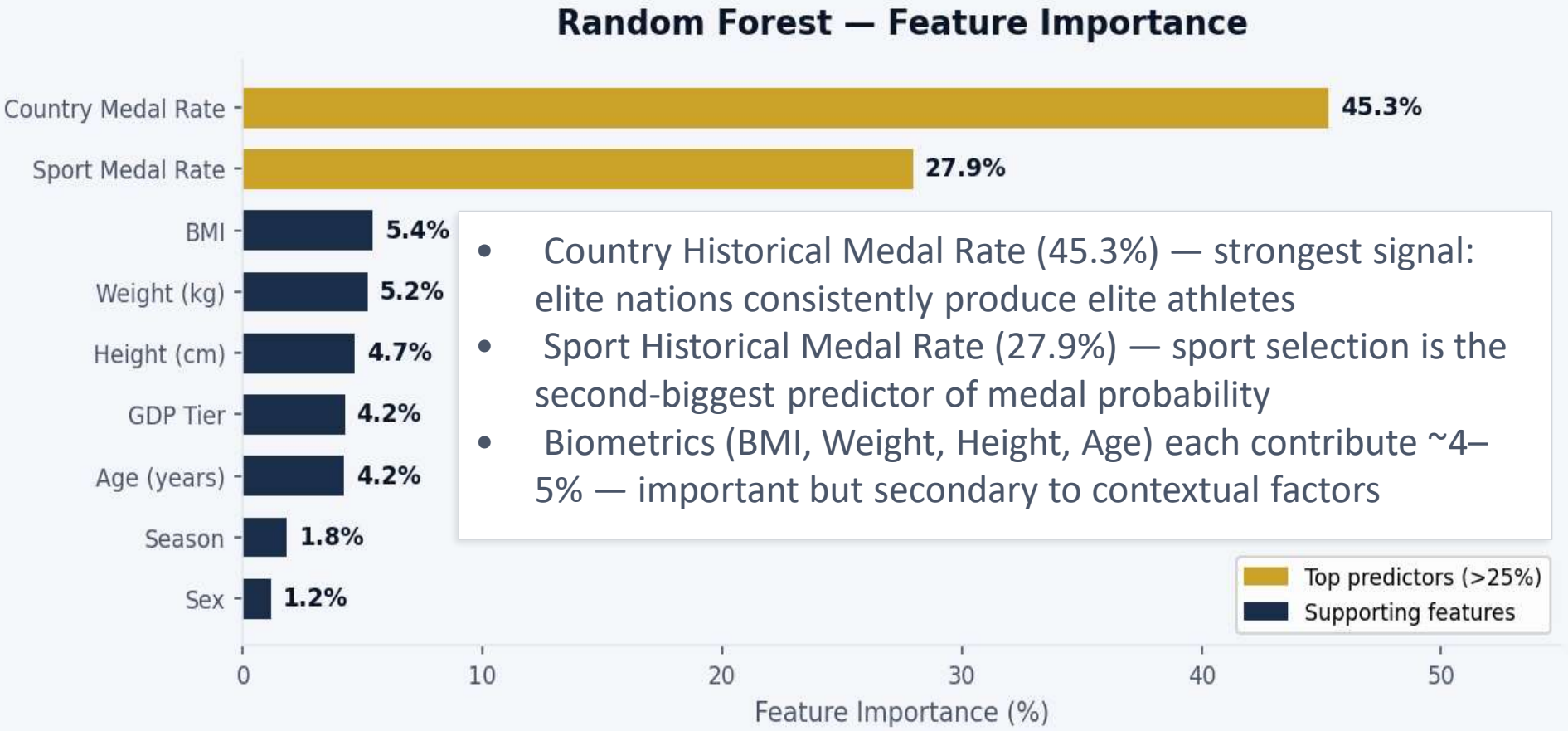
Q: Is it a population problem (too many, too diverse)?
A: No — China has a comparable scale and leads the world. India's problem is lack of targeted talent identification and development pipelines, not population scale.

Q: Is it a sports culture issue?
A: Yes — cricket absorbs 80%+ of all corporate sponsorship, media attention and youth aspiration. Olympic sports operate in a funding vacuum with minimal public visibility.

Q: What does the medal rate reveal?
A: India's 0.006 medal rate (6 medals per 1,000 athletes) is among the world's worst. Nations with smaller delegations convert a far higher % of participants to medal winners.

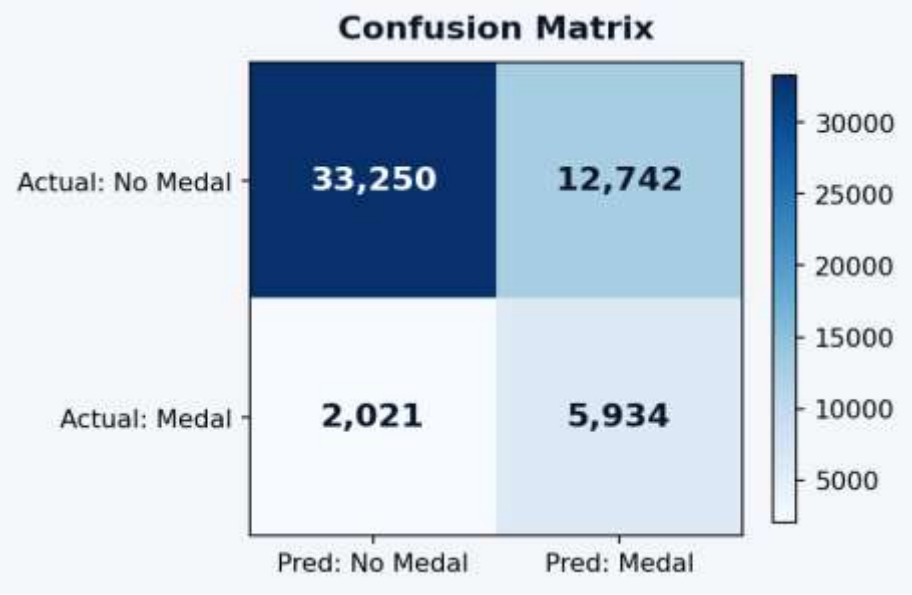
Q: What is the structural root cause?
A: Post-1980 collapse of hockey dominance was never compensated by investment in other sports. No systematic state sports model (unlike China, Cuba or Kenya) exists in India.

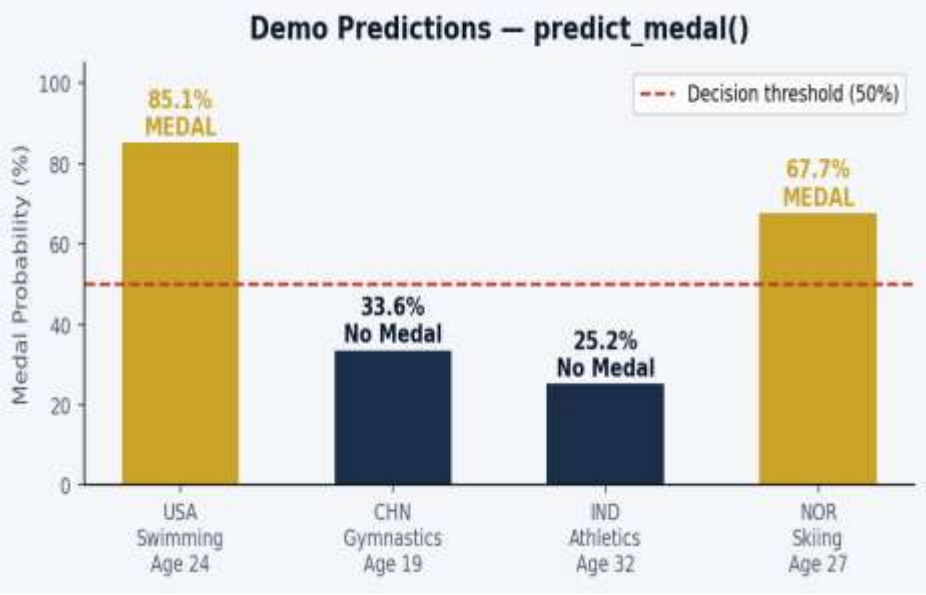
07 | Medal Prediction Model



- Country Historical Medal Rate (45.3%) — strongest signal: elite nations consistently produce elite athletes
- Sport Historical Medal Rate (27.9%) — sport selection is the second-biggest predictor of medal probability
- Biometrics (BMI, Weight, Height, Age) each contribute ~4–5% — important but secondary to contextual factors

07 | Medal Prediction Model





✅ **True Negatives: 33,250**
Correctly predicted No Medal-72% of actual non-winners identified

✅ **True Positives: 5,934**
Correctly predicted Medal — 74.6% of actual medal winners caught

⚠ **False Positives: 12,742**
Predicted Medal but got none

⚠ **False Negatives: 2,021**
Missed winners — only 25.4% of winners slipped through

08 | Key Conclusions

💰 GDP Drives Medals

Strong positive correlation between GDP per capita and all-time medal count. Wealthy nations invest more in athlete development systems.

⚖️ Quality > Scale

Small high-income nations (Scandinavia, Eastern Europe) show higher medal rates per athlete than large but lower-income countries.

👩‍🦰 Gender Parity Progress

Female participation rose from near-zero in 1896 to ~45% by 2016, driven by IOC policy reforms. Still ~5% short of parity.

🏆 USA's Sustained Dominance

The USA leads all-time Summer Olympic medals, sustained by private sports infrastructure, college athletics and diverse talent pipelines.

CN China's Rapid Rise

Post-1984 investment in targeted high-medal sports (diving, gymnastics, weightlifting) made China the #2 Summer power by 2008 Beijing.

IN India's Chronic Gap

Despite 1.4B population, India has won only 35 Olympics medals — 71% from hockey alone. Medal rate is among the world's lowest.

📏 Physical Profiles Matter

BMI and sport match significantly affect medal probability. Sports-specific training programs need physical profiling from youth level.

📉 Post-1980 Hockey Collapse

India's switch from natural grass to artificial turf in 1980 eliminated its key competitive advantage, causing a multi-decade medal drought.

09 | Policy Implications

1. **Sports Budget Increase**
   Raise sports ministry allocation to 0.5–1% of GDP. Benchmark: China spends ~$500M annually on elite athlete programs alone.

2. **TOPS Expansion**
   Scale up Target Olympic Podium Scheme (TOPS) from ~250 to 2,500 athletes. Include grassroots identification, not just established names.

3. **Sport Diversification**
   Reduce dependence on Hockey. Invest heavily in Shooting, Wrestling, Badminton, Athletics & Boxing where India shows emerging talent.

4. **Gender Inclusion**
   Mandate 40% female participation funding in all national sports bodies. India's female medal share is below even global developing-nation averages.

09 | Policy Implications

5

Physical Profiling Programs

Institute nationwide BMI and physical attribute profiling at school level to identify and route children to sports aligned with their physique.

6

Corporate Sponsorship Incentives

Offer CSR tax incentives for corporate sports sponsorship. Create public-private athlete academies on the Kenya / Jamaica community model.

7

Infrastructure Modernization

Build sport-specific modern training centers in Tier 2/3 cities. India loses talent because athletes cannot afford to relocate to metro facilities.

Thank You

Key Takeaway: India punches far below its weight at the Olympics — a solvable problem with targeted investment, sport diversification & athlete-centric policy reform.